

CODE

```
class Student {  
    String name;  
    int rollNumber;  
    double gpa;  
  
    public Student(String name, int rollNumber, double gpa) {  
        this.name = name;  
        this.rollNumber = rollNumber;  
        this.gpa = gpa;  
    }  
}
```

```
class StudentTest {  
  
    public static Student getTopStudent(Student[] students) {  
        Student top = students[0];  
  
        for (int i = 1; i < students.length; i++) {  
            if (students[i].gpa > top.gpa) {  
                top = students[i];  
            }  
        }  
        return top;  
    }  
}
```

```

public static void main(String[] args) {
    Student[] s = new Student[5];

    s[0] = new Student("Ali", 1, 3.2);
    s[1] = new Student("Ahmed", 2, 3.8);
    s[2] = new Student("Sara", 3, 3.5);
    s[3] = new Student("Zain", 4, 3.9);
    s[4] = new Student("Ayesha", 5, 3.6);

    Student top = getTopStudent(s);

    System.out.println("Top Student: " + top.name + " GPA: " +
top.gpa);
    }
}

import java.util.ArrayList;

public class DuplicateRemoval {

    public static int[] removeDuplicates(int[] arr) {
        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < arr.length; i++) {
            if (!list.contains(arr[i])) {
                list.add(arr[i]);
            }
        }
    }
}

```

```

        int[] result = new int[list.size()];
        for (int i = 0; i < list.size(); i++) {
            result[i] = list.get(i);
        }

        return result;
    }
}

class Book {
    String title;
    String author;
    double price;

    public Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }
}

public class BookTest {

    public static void expensiveBooks(Book[] books) {
        for (int i = 0; i < books.length; i++) {
            if (books[i].price > 1000) {
                System.out.println(books[i].title + " - " + books[i].price);
            }
        }
    }
}

```

```
    }  
  }  
}
```

```
public static void main(String[] args) {
```

```
    Book[] b = new Book[5];
```

```
    b[0] = new Book("Java", "Author1", 1200);
```

```
    b[1] = new Book("C++", "Author2", 800);
```

```
    b[2] = new Book("Python", "Author3", 1500);
```

```
    b[3] = new Book("AI", "Author4", 2000);
```

```
    b[4] = new Book("DBMS", "Author5", 900);
```

```
    expensiveBooks(b);
```

```
  }
```

```
}
```

```
class Employee {
```

```
    int id;
```

```
    String name;
```

```
    double salary;
```

```
    public Employee(int id, String name, double salary) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        this.salary = salary;
```

```
    }
```

```
}
```

```
class EmployeeTest {

    public static void updateSalary(Employee[] emp) {
        for (int i = 0; i < emp.length; i++) {
            if (emp[i].salary < 50000) {
                emp[i].salary = emp[i].salary * 1.10;
            }
        }
    }

    public static void main(String[] args) {
        Employee[] e = new Employee[5];

        e[0] = new Employee(1, "Ali", 40000);
        e[1] = new Employee(2, "Ahmed", 55000);
        e[2] = new Employee(3, "Sara", 30000);
        e[3] = new Employee(4, "Zain", 60000);
        e[4] = new Employee(5, "Ayesha", 45000);

        updateSalary(e);

        for (Employee emp : e) {
            System.out.println(emp.name + " New Salary: " + emp.salary);
        }
    }
}
```

```
public class GradeAdjustment {  
  
    public static int[] adjustMarks(int[] marks) {  
        int[] newMarks = new int[marks.length];  
  
        for (int i = 0; i < marks.length; i++) {  
            newMarks[i] = marks[i] + 5;  
  
            if (newMarks[i] > 100) {  
                newMarks[i] = 100;  
            }  
        }  
  
        return newMarks;  
    }  
}
```

```
public class FrequencyArray {  
  
    public static int[] getFrequency(int[] arr) {  
  
        int max = 0;  
  
        for (int i = 0; i < arr.length; i++) {  
            if (arr[i] > max) {  
                max = arr[i];  
            }  
        }  
    }  
}
```

```
    }  
  
    int[] freq = new int[max + 1];  
  
    for (int i = 0; i < arr.length; i++) {  
        freq[arr[i]]++;  
    }  
  
    return freq;  
}  
}
```

OUTPUT

```
C:\Users\hussa\.jdk\openjdk-24.0.2+12-54\bin\java.exe "-java  
Java - 1200.0  
Python - 1500.0  
AI - 2000.0  
  
Process finished with exit code 0
```